UNRAMIFIED EXTENSIONS COMPOSE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ramification indices are multiplicative in towers of discrete valuation rings. invoked at proof lines 33150-33151. Combines e = 1 at each step to e = 1 for M/K.

Source reference. Original lemma 0BRL: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32952–32952. Cited at proof lines 33150-33151 to compose the ramification indices; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K. Let $M/L/K$ be finite separable extensions. Let B be the integral closure of A in L. If L/K is unramified with respect to A and M/L is unramified with respect to $B_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ of B, then M/K is unramified with respect to A.*

Proof. Let C be the integral closure of A in M. Every maximal ideal $\mathfrak{m}'$ of C lies over a maximal ideal $\mathfrak{m}$ of B. Then the lemma follows from the multiplicativity of ramification indices (Lemma 0BRL) and the fact that we have the tower $\kappa(\mathfrak{m}')/\kappa(\mathfrak{m})/\kappa_A$ of finite extensions of fields. □

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